

Prabesh Pandey

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EDUCATION

Masters in Applied Geography

2024 – 2026

Texas State University, San Marcos, Texas

Major: GIS and Cartography

- Geographic Application of Remote Sensing
- LiDAR and SFM Data Processing and Analysis
- Technical GIS
- GPS and GIS
- GIS II – Introduction to Python
- Multivariate Quantitative Analysis

Bachelor's Degree in Geology

2017 – 2022

Tribhuvan University, Nepal | Major: Geology | Minor: Botany and Zoology

- Geology of the Himalayas
- Stratigraphy, Petrology, Structural Geology
- Geophysics and Geochemistry, Geomorphology
- Crystallography and Mineralogy
- Exploration and Mining Geology, Engineering Geology
- Hydrogeology, Historical Geology

GIS AND REMOTE SENSING PROJECT EXPERIENCE

- Georeferenced crime locations and applied buffer analysis on the San Antonio highway road network in ArcGIS Pro, quantifying crime within a 1-mile buffer zone.
- Used LiDAR point cloud to analyze patterns of gentrification in Austin, Texas through spatial-temporal land cover change detection analysis.
- Used Landsat satellite imagery to analyze temporal shifts of the Melamchi flood tragedy in Nepal and created a risk analysis buffer to prevent loss of lives.
- Integrated temperature, rainfall, and groundwater data over a decade in ArcGIS Pro and built a Python toolbox to identify wildfire-prone areas.
- Built a network analysis tool in ArcGIS to quantify travel time and distance to the nearest emergency healthcare facilities in Hays County, analyzing demand and coverage with respect to population (2SFCA methodology).
- Used the kriging interpolation method and converted interpolated raster to vector to analyze wind direction and speed in Hays County.
- Manually collected GPS data for the Disc Golf Course at Texas State University using a Bad Elf device; built a Web Map using Experience Builder and a static map using ArcGIS Pro.
- Used Landsat imagery and supervised classification to analyze land cover change and vegetation indices using Google Earth Engine (GEE) and ERDAS Imagine.
- Developed a least-cost path analysis using railroad networks and nuclear facility locations to optimize waste transport routes while minimizing population exposure.

FIELD EXPERIENCE

Limestone Quarry Site Investigation	2022
Reserve Estimation of Iron Deposits	2022
Groundwater Distribution and Exploitation	2022
Geological Compass Traverse	2022
Study of Detail Litho-Stratigraphy in Nepal	2022
Engineering and Geological Route Mapping	2021
Mapping Flood and Landslide Hazard Risk Zone	2021
Reserve Estimation of Mineral Resources	2021
Field Identification of Geological and Geo-morphological Features	2020
Study of Detail Lithostratigraphy of Central Nepal	2019

SKILLS

GIS and Cartography: Network analysis, interpolation methods, georeferencing, raster/vector analysis, cost-path analysis, buffer analysis, Model Builder, cartographic design, interactive maps.

Remote Sensing: Image statistics, geometric correction, mosaicking, radiometric correction, vegetation indices, principal component analysis, tassal cap transformation.

LiDAR: Preprocessing and post-processing point cloud data, QA/QC, accuracy assessment, terrain analysis and visualization.

GPS: Data collection, surveying, processing, QA/QC, schema building, device configuration.

Statistical Analysis: ANOVA, Pearson correlation, regression analysis, multivariate statistics.

SOFTWARE AND TOOLS

ArcGIS Pro, ArcGIS Online, ERDAS Imagine, LP360, Field Map Designer, Instant App, Experience Builder, Jupyter Notebook, Python, Cartopy, Scrapy, Google Earth Engine (GEE), SQL, JMP, Microsoft Office

CERTIFICATIONS AND TRAINING

Cartography Certificate	Oct 2025
Python for Everyone – Coursera	Feb 2025
Data Visualization Certificate	2025
Database and SQL Certificate	2025
CITI Training – Human Subjects Protection	2025

PROFESSIONAL EXPERIENCE

Graduate Instructional Assistant – Physical Geology

Fall 2024 – Present

Texas State University | Teach physical geology lab; conduct exercises in rock/mineral identification, earthquake science, hydrogeology, stratigraphy, and geological cross-sections.

LEADERSHIP

Masters Faculty Representative – Graduate Forum

Fall 2025 – Present

PRESENTATIONS AND CONFERENCES

Graduate Student Research Conference (GSRC) – Poster Presentation	2026
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PANGEA Conference Presentation – Organized Crime and Environmental Harm in Central America2025
SEG Seminars: Landslide Mitigation, Climate Change, Earthquake Mitigation, Groundwater Investigation, MCT
and Inverted Metamorphism in Himalaya 2021–2022

AWARDS AND FELLOWSHIPS

Gildea Graduate Academic Excellence Scholarship in Geography (SG6427)	2026
PANGEA Research Conference Award	2026
TSUS Graduate Endowed Fellowship in Liberal Arts	Fall 2025 & Spring 2025
Graduate Merit Scholarship	Fall 2024 & Spring 2025

PUBLICATIONS (IN PROGRESS)

- Organized Crime and Environmental Harm: A Case Study of Guatemala and Panama
- Drug Trafficking, Criminality, and Violence in Central America

LANGUAGES

English (Professional) | Nepali (Native) | Hindi (Professional)